UK Rental Price Indicators: A Regional Spatiotemporal Analysis

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1 Data Collection and Preprocessing

The ingestion pipeline combines ONS PIPR monthly average rents with 10.18 million mapped HM Land Registry Price Paid transactions, 59.25 million mapped `police.uk` street-crime records, and ONS local-authority gross disposable household income (GDHI) and population. The August 2024 ONS Postcode Directory maps transaction postcodes and crime LSOAs to a common LAD code. PIPR and the lookup tables use different LAD codes for Barnsley and Sheffield; these are reconciled using area names and regions. The rent and event window is 2015–2024; annual GDHI is joined from the previous year.

PySpark checks dates, positive sale prices, required identifiers and geographic matches before storing accepted events as Parquet and rejected rows with reason codes. Records without a reliable geographic match are kept out of the monthly aggregates. Monthly sale counts and medians use standard-market transactions only (category A, record status A); sale prices are contextual housing-market indicators rather than rental observations. A crime LSOA is assigned only when at least 95% of supporting postcodes agree on the LAD. Where a police-force source file is absent, crime is left null rather than coded as zero. The resulting panel has one published area-level rent estimate per LAD–month (316 LADs, 120 months), not individual tenancies. Independent checks confirm unique keys, complete rent and income, and agreement between CSV and Parquet outputs; processing metadata records coverage and rejection counts.

2 Feature Engineering

We output a balanced LAD×month panel (316 authorities, 120 months, 37,920 rows) as `features.csv` (120 variables) with `feature_dictionary.csv` marking association vs. forecast use and temporal role (contemporaneous, lagged, quality flag).

2.1 Data quality treatment

`police.uk` street crime is incomplete. **Partial uploads** (force month <50% of its median volume, or LAD <25%) are flagged and excluded (32 LAD-months). **Short gaps** (≤12 consecutive months) are seasonally imputed (value at $t-12$ × regional YoY ratio; 176 rows, `crime_imputed`; LOYO median APE 6.9%). **Long gaps** (Greater Manchester, 2019-07–2024-12, 660 rows) stay null. **Home Office** recorded crime (CSP→LAD mapping, population splits, unassigned offences redistributed from prior-quarter CSP shares) is complete on all rows. Income and population use year $Y-1$; all rolls are trailing.

Table 1: Street-crime quality summary.

Issue	Treatment	Count
Partial upload	Exclude + flag	32
Gap ≤12 mo.	Seasonal impute	176
GMP long gap	Null; use HO features	660

2.2 Feature construction

PySpark joins rent, geography (centroid, urban share, deprivation, 5-neighbour graph), housing rolls/mix, calendar shocks, and two crime systems: monthly street rates (12-mo. rolls need ≥6 valid months) and quarterly recorded crime (4-quarter sums per 1,000 residents). Neighbour means share the same month.

3 Analysis and Results

Space and time. Using PySpark on `features.csv`, the average LAD's nominal rent rose 33% over 2016–24 (Fig. 1a), fastest in regional cities (Salford +61%, Leicester +57%, Manchester +55%) and slowest in Hartlepool, Blackpool and Ceredigion (≈16%). London was flat in 2017–21 and grew only 32%, behind every region but the North East (Fig. 1b).

Within-area association. On a LAD×year panel (2,844 rows; annual changes avoid overlapping 12-month windows) we regress log rent change on log changes in median sale price and recorded crime and on prior-year income and population growth. LAD and year fixed effects are removed by demeaning with Spark window functions before `LinearRegression`; SEs are clustered by LAD (Table 2). Prices look unrelated to rents until year effects are removed: nationally they moved apart (2023–24 rents +6–8%/yr while prices stalled), hiding a local link where 10% faster price growth goes with 1.2% faster rent growth. Crime shows no within-area association; within R^2=0.10.

Table 2: Annual LAD rent growth on local changes; LAD-clustered SE. $^{**}p<.01$, $^{***}p<.001$.

	Pooled	LAD FE	LAD + year FE
House price (dlog)	0.025^{**} (0.010)	0.007 (0.011)	$\mathbf{0.117}^{***}$ (0.013)
Recorded crime (dlog)	-0.011^{**} (0.004)	−0.007 (0.004)	0.007 (0.004)
GDHI/head growth ($Y-1$)	0.380^{***} (0.016)	0.354^{***} (0.022)	0.069^{***} (0.021)
Population growth ($Y-1$)	1.264^{***} (0.068)	1.771^{***} (0.094)	0.682^{***} (0.097)
R^2 (within for FE)	0.278	0.323	0.101

Market profiles. KMeans clusters the 316 LADs on five standardised profiles (log rent, rent growth, price-to-income, log crime, urban share; GDHI omitted as it correlates 0.81 with rent). Silhouette ties at k=3 (0.482) and k=4 (0.480); k=4 splits urban LADs into stagnant former industrial areas (C0, £580, +28%) and fast-rising regional cities (C2, +42%), beside rural LADs (C1) and London with its commuter belt (C3, £1,382, +28%; Fig. 1c).

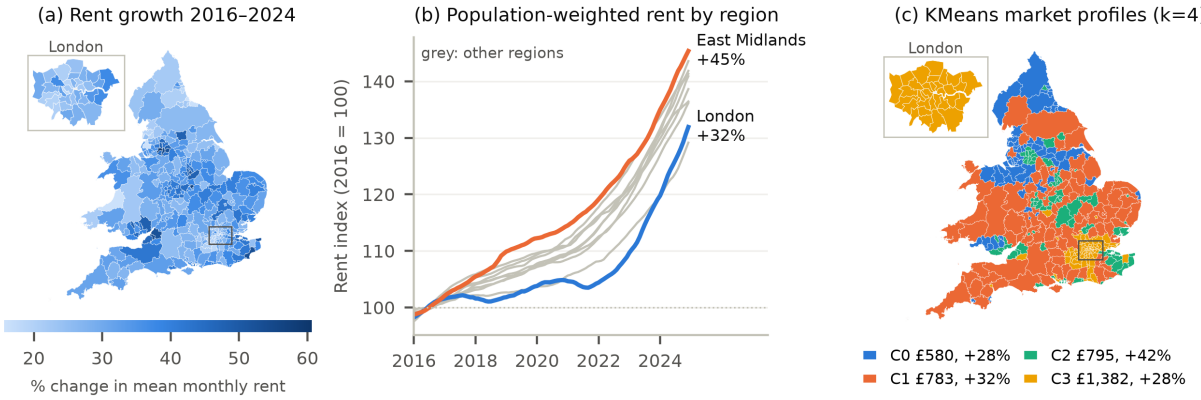


Figure 1: (a) Rent growth by LAD; (b) regional rent index; (c) KMeans profiles (mean rent, growth).

Limitations and running. Results are associations on nominal rents; recorded crime partly reflects recording practice. To run: `uv sync`, the pipeline in `README.md`, then `analysis/0[1-4]_*.py`.